

Robin Thibaut

Head of Agentic Systems R&D at Zanskar

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1. Appointments

Zanskar Geothermal & Minerals

Head of Agentic Systems R&D

August 2026–Present

- Lead research and development on agent systems for long-running scientific, technical, and operational work.
- Scope: problem decomposition, tool and data use, context management over extended tasks, uncertainty handling, and evaluation across multiple steps.

Senior Computational Geoscientist

October 2024–August 2026

- Owned the subsurface-modeling codebase for 3D geological and thermal models used in geothermal exploration.
- Built data pipelines and scientific-software workflows for fault geometry, stratigraphy, and finite-element thermal simulation.
- Calibrated models against well and geophysical data; quantified uncertainty and used optimization to support well targeting.

Postdoctoral Researcher — Lawrence Berkeley National Laboratory

August 2023–September 2024

Berkeley; Watershed Function Scientific Focus Area

- Model–data integration for mountainous watersheds combining geophysical observations, hydrologic modeling, and machine learning.

Postdoctoral Researcher, TURBEAMS — Ghent University

March 2023–May 2023

- Applied machine learning to multibeam echosounder data to estimate turbidity and suspended particulate matter in coastal waters.

PhD Fellow — Ghent University

March 2019–March 2023

- Doctoral research on Bayesian Evidential Learning, experimental design, and uncertainty reduction for subsurface problems, including wellhead protection and temperature monitoring.
- Co-developed SKBEL, open-source research software for Bayesian Evidential Learning (PhD-era).

Project Engineer — G-tec

March 2018–January 2019

- Worked across planning, mobilization, acquisition, processing, and reporting for marine geophysical surveys: unexploded-ordnance detection, seabed-layer characterization, bathymetry, and hydrographic investigation.
- Magnetometry, seismic refraction, side-scan sonar, and multibeam methods.

Geophysics Intern — Institute of Technology of Cambodia, Monduliri Province, Cambodia

- Time-domain induced polarization and electrical resistivity for gold exploration: survey design, acquisition, processing, and inversion.

2. Education

PhD, Geological Sciences — Ghent University

2019–2023

Dissertation: *Machine Learning for Bayesian Experimental Design in the Subsurface*

MSc, Geological Engineering — University of Liège

2014–2017

BSc, Geological Sciences — Université libre de Bruxelles

2010–2014

Foundation: field observation, mapping, and laboratory practice.

3. Journal Articles

- Zhang, L., Daniilidis, A., Dieudonné, A.-C., **Thibaut, R.**, & Hermans, T. (2026). A Bayesian Evidential Learning Framework for Safety and Performance Prediction in Thermo-Hydro-Mechanical Coupled Deep Mine Geothermal Systems. *Journal of Rock Mechanics and Geotechnical Engineering*. doi:10.1016/j.jrmge.2025.11.022.
- Chen, H., **Thibaut, R.**, Chou, C., Xiong, C., & Wu, Y. (2026). A ModEx Framework for Watershed Subsurface Investigation with Limited Geophysical Data Using Machine Learning and Hydrologic Modeling. *Geophysical Research Letters*, 53(2), e2025GL119953. doi:10.1029/2025GL119953.
- Wang, L., Xu, Z., Wang, C., **Thibaut, R.**, Ulrich, C., Sprenger, M., Uhlemann, S., Wu, Y., King, E., Wainwright, H., Carroll, R. W. H., Beutler, C., Williams, K. H., & Dafflon, B. (2025). The Role of Snowmelt and Subsurface Heterogeneity in Headwater Hydrology of a Mountainous Catchment in Colorado: A Model–Data Integration Approach. *Water Resources Research*, 61(10), e2025WR040651. doi:10.1029/2025WR040651.
- Tas, L., Hartog, N., Bloemendal, M., Simpson, D., Robert, T., **Thibaut, R.**, Zhang, L., & Hermans, T. (2025). Efficiency and heat transport processes of low-temperature aquifer thermal energy storage systems: new insights from global sensitivity analyses. *Geothermal Energy*, 13, Article 2. doi:10.1186/s40517-024-00326-1.
- Zhang, L., Dieudonné, A.-C., Daniilidis, A., Dong, L., Cao, W., **Thibaut, R.**, Tas, L., & Hermans, T. (2025). Thermo-Hydro-Mechanical Modeling of Geothermal Energy Systems in Deep Mines: Uncertainty Quantification and Design Optimization. *Applied Energy*, 377, 124531. doi:10.1016/j.apenergy.2024.124531.
- Thibaut, R.**, Compaire, N., Lesparre, N., Ramgraber, M., Laloy, E., & Hermans, T. (2022). Comparing Well and Geophysical Data for Temperature Monitoring Within a Bayesian Experimental Design Framework. *Water Resources Research*, 58(11). doi:10.1029/2022WR033045.
- Thibaut, R.**, Laloy, E., & Hermans, T. (2021). A new framework for experimental design using Bayesian Evidential Learning: The case of wellhead protection area. *Journal of Hydrology*, 603, 126903. doi:10.1016/j.jhydrol.2021.126903.
- Diep Cong-Thi, Linh Pham Dieu, **Thibaut, R.**, Paepen, M., Ho, H. H., Nguyen, F., & Hermans, T. (2021). Imaging the Structure and the Saltwater Intrusion Extent of the Luy River Coastal Aquifer (Binh Thuan, Vietnam) Using Electrical Resistivity Tomography. *Water*, 13(13), 1743. doi:10.3390/w13131743.
- Thibaut, R.**, Kremer, T., Royen, A., Ngun, B. K., Nguyen, F., & Hermans, T. (2021). A new workflow to incorporate prior information in minimum gradient support (MGS) inversion of electrical resistivity and induced polarization data. *Journal of Applied Geophysics*, 187, 104286. doi:10.1016/j.jappgeo.2021.104286.

4. Dissertation

Thibaut, R. (2023). *Machine Learning for Bayesian Experimental Design in the Subsurface*. PhD dissertation, Ghent University.

5. Datasets

- Ifrene, G., Rabade, S., Pollack, A., **Thibaut, R.**, Liang, Z., Yu, J., & Zheng, F. (2026). Lightning Dock Geothermal Field Data Package, Hidalgo County, New Mexico [Data set]. Geothermal Data Repository. Zanskar geothermal & minerals inc. <https://gdr.openei.org/submissions/1856>.
- Liang, Z., Brisson, S. C., **Thibaut, R.**, Yu, J., Hoiland, C., & Pollack, A. (2024). Zanskar’s GeoGym [Dataset]. Kaggle. <https://www.kaggle.com/datasets/robinthibautzanskar/zanskars-geogym>.
- Lesparre, N., Compaire, N., Hermans, T., & **Thibaut, R.** (2022). 4D Temperature Monitoring [Dataset]. Kaggle. <https://www.kaggle.com/dsv/3819983>.
- Thibaut, R.** (2021). WHPA Prediction [Dataset]. Kaggle. <https://www.kaggle.com/dsv/2648718>.

6. Conference Contributions

- Liang, Z., Yu, J., **Thibaut, R.**, Zheng, F., Hoiland, C., & Pollack, A. (2026). Surrogate Modeling for Geothermal Systems: Accelerating Optimization, History Matching, and Uncertainty Quantification. *Proceedings, 51st Workshop on Geothermal Reservoir Engineering*, 1–10.
- Zheng, F., **Thibaut, R.**, Liang, Z., Yu, J., & Pollack, A. (2026). Thermal Gradient Signatures of Fault-Controlled Hydrothermal Systems: Insights from Conceptual Models and Field Data. *Proceedings, 51st Workshop on Geothermal Reservoir Engineering*, 1–8.
- Liang, Z., Brisson, S. C., **Thibaut, R.**, Yu, J., Hoiland, C., & Pollack, A. (2025). GeoGym: a Benchmark Dataset for Evaluating Geothermal Exploration Strategies. *Proceedings, 50th Workshop on Geothermal Reservoir Engineering*, 1–7.

Wu, Y., Balan, K., Chou, C., Chung, J. W., Dafflon, B., Falco, N., Panaro, J., Persaud, A., Quiter, B., Rofors, E., **Thibaut, R.**, Wang, J., Whittaker, M. L., & Wu, J. (2024). RARE EARTH ELEMENTS – MULTIPHYSICS AI-AIDED AUTONOMOUS PROSPECTING (REE-MAP). *Geological Society of America Abstracts with Programs*, 56(5), 1. doi:10.1130/abs/2024AM-404345.

Thibaut, R., Ferré, T., & Hermans, T. (2023). Sequential Optimization Of Temperature Measurements To Estimate Groundwater Surface Water Interactions. *AGU Fall Meeting 2023, Abstracts*, 1.

Thibaut, R., Compaire, N., Lesparre, N., Ramgraber, M., Laloy, E., & Hermans, T. (2022). Comparing well and geophysical data for temperature monitoring within a Bayesian Experimental Design framework. *Computational Methods in Water Resources, Abstracts*, 20.

Thibaut, R., Hermans, T., & Laloy, E. (2021). A new framework for experimental design using Bayesian Evidential Learning: the case of wellhead protection area. *AGU Fall Meeting 2021, Abstracts*, 1.

Thibaut, R., Hermans, T., & Laloy, E. (2021). Bayesian Evidential Learning combined with experimental design: the case of wellhead protection area prediction. *48th IAH Congress, Abstracts*.

Hermans, T., Compaire, N., **Thibaut, R.**, & Lesparre, N. (2021). Bayesian evidential learning: an alternative to hydrogeophysical coupled inversion. *First International Meeting for Applied Geoscience & Energy Expanded Abstracts*, 3125–3129. doi:10.1190/segam2021-3580979.1.

Thibaut, R., Hermans, T., & Laloy, E. (2020). A new framework to reduce uncertainty in Wellhead Protection Area prediction using Bayesian Evidential Learning. *10th International Congress on Environmental Modelling and Software (iEMSs 2020), Abstracts*, 1.

Thibaut, R., Kremer, T., Royen, A., Ngun, B. K., Nguyen, F., & Hermans, T. (2019). A new approach to incorporate prior information in MGS inversion of ERT/IP data. *Near Surface Geoscience Conference and Exhibition 2019*, 5. doi:10.3997/2214-4609.201902391.

7. Research Software

Thibaut, R. & Ramgraber, M. (2021). SKBEL — Bayesian Evidential Learning framework built on top of scikit-learn [Software, PhD-era research software]. Zenodo, v2.0.0. doi:10.5281/zenodo.6205242.

Deleersnyder, W. & **Thibaut, R.** (2022). Scale-dependent wavelet-based regularization scheme for geophysical 1D inversion [Software]. Zenodo. doi:10.5281/zenodo.6552695.

Thibaut, R. & Vandekerckhove, G. (2021). pysgems — Use SGeMS (Stanford Geostatistical Modeling Software) within Python [Software, deprecated]. Zenodo, v1.3. doi:10.5281/zenodo.4773587.

8. Peer Review Service

Water Resources Research; Geophysical Research Letters; Journal of Hydrology; Advances in Water Resources; Geophysical Journal International; GEOPHYSICS; Pure and Applied Geophysics; Hydrogeology Journal; GEUS Bulletin (2021–2024).

9. Teaching

Teaching Assistant — Groundwater Modeling, Ghent University | 2019–2023

Organizer and Speaker — Python Workshop, Ghent University | March 2022

Private Tutor — Mathematics and Physics for university students | 2016–2019

10. Languages

English (Fluent); French (Fluent); Dutch (Intermediate); Vietnamese (Intermediate).